

Testing and Correcting for Insufficient Follow-Up in Covariate-Dependent Cure Models: From Categorical Tests to a Heavy-Tailed Extrapolation, and Back

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Abstract

Cure models are widely used to analyze time-to-event data in which a subpopulation never experiences the event of interest: for example, cancer patients who are effectively cured, or borrowers who will never default. A central practical difficulty in fitting these models is *insufficient follow-up*: if the observation window is too short relative to the tail of the event-time distribution, the cured fraction is systematically overestimated, because patients who would eventually experience the event but have not yet done so are indistinguishable, within the observed window, from those who are truly cured. Extreme value theory (EVT) offers a principled way to relax the traditional “sufficient follow-up” assumption by modeling the tail of the susceptible population’s survival distribution directly. This report documents an applied project built around four recent papers on this problem, combining (i) a statistical test for whether follow-up is long enough, (ii) an EVT-based correction to the cure-rate estimator when it is not, and (iii) an investigation of how both tools behave as the number of covariates grows. All methods are applied to the real `survival::colon` dataset (929 colon-cancer patients, two endpoints), rather than to synthetic data alone, and several implementation bugs in both a public reference package and our own from-scratch code were found and corrected along the way. The project was built as a demonstration piece to accompany a PhD application to VU Amsterdam on exactly this topic.

Contents

1	Motivation and Problem Statement	2
2	Literature Anchors	2
3	Data	3
4	Phase 1: Baseline Replication of the Sufficient Follow-Up Tests	3
4.1	Method	3
4.2	Implementation issues found and fixed	4
4.3	Results	4
5	Phase 2: Extrapolation Estimator and the Curse of Dimensionality	5
5.1	Method	5
5.2	Validation and bugs found	5
5.3	Result: recurrence vs. death, and triangulation with Phase 1	5
5.4	Curse-of-dimensionality demonstration	6
5.5	Comparison against a gradient-boosted alternative (gbex)	7

6 Phase 5: Simulation Study	8
6.1 Part A: Bias–variance tradeoff as follow-up length varies	8
6.2 Part B: Method 1 vs. Method 2 as the number of covariate categories grows	9
7 Discussion: What This Project Demonstrates	9
8 Future Work	10

1 Motivation and Problem Statement

Standard survival analysis assumes that, given enough time, every subject in the study population will eventually experience the event of interest. *Cure models* relax this by positing a latent split of the population into a *susceptible* fraction, who will eventually experience the event, and a *cured* (or immune) fraction, who never will. This structure is natural in oncology (a meaningful fraction of treated cancer patients never relapse) and in credit risk (a meaningful fraction of borrowers never default over the life of a loan).

Estimating the cured fraction from data requires distinguishing, among subjects who have not yet had the event by the end of the study, those who never will from those who simply have not yet reached their (long) event time. This is only identifiable if the study’s follow-up is long enough that essentially all susceptible subjects have already had the event by the end of observation: the *sufficient follow-up* (SFU) assumption. When follow-up is insufficient, naive nonparametric estimators (e.g. Kaplan–Meier-based cure-rate estimates) are biased: they systematically *overestimate* the cured fraction, because long-surviving susceptible subjects are counted as if cured.

The applied problem this project addresses has two parts:

1. **Detection.** Given a dataset, is follow-up actually long enough to trust a naive cure-rate estimate, overall and within covariate subgroups?
2. **Correction.** If not, can the tail behavior of the event-time distribution be modeled well enough, using extreme value theory, to correct the naive estimate, and does this correction remain usable as more covariates are added?

This project was built around four recent papers that address exactly these two questions from different angles, and applies all of them, end to end, to a single real dataset, rather than treating each paper as an isolated synthetic-data exercise.

2 Literature Anchors

Four papers anchor the project. All four were read in full and their key derivations were worked through by hand before any code was written.

Yuen & Musta (2024/2026)

(Yuen and Musta, 2024) An unconditional (marginal, no covariates) hypothesis test for sufficient follow-up, based on a smoothed Grenander-type estimator of the boundary of the event-time distribution’s support. This is the baseline detection tool.

Yuen, Musta & Van Keilegom (2025)

(Yuen et al., 2025) Extends the above test to categorical covariates. Proposes two complementary procedures: *Method 1*, an intersection-union test (IUT) that only declares “sufficient follow-up for all subgroups” if every subgroup individually clears significance; and *Method 2*, a bootstrap procedure that identifies which specific subgroup is most likely to have insufficient follow-up.

Xie, Escobar-Bach & Van Keilegom (2023/2024)

(Xie et al., 2023) A sufficient-follow-up test in the Gumbel maximum-domain-of-attraction case, using an Aitken- Δ^2 -style correction. Uses the opposite null-hypothesis convention to Yuen & Musta (H_0 : sufficient, rather than H_0 : insufficient), which matters when combining results across papers.

Escobar-Bach & Van Keilegom (2019/2023)

(Escobar-Bach and Van Keilegom, 2019) The *correction* paper, and the anchor for the extrapolation component of this project. In the Fréchet maximum-domain-of-attraction (heavy-tailed) case, it constructs a Beran-type conditional Kaplan–Meier estimator, then corrects its tail using a Hill-type extreme-value index estimator $\hat{\gamma}(x)$ and an extrapolated cure-probability estimator $\hat{p}(x)$, allowing the cure probability to be estimated *beyond* the observed follow-up horizon.

3 Data

All methods are applied to `survival::colon`, a public dataset of 929 patients from a colon cancer clinical trial, with two competing endpoints: *recurrence* and *death* (all-cause mortality). Covariates used across the project include age, sex, treatment arm, number of positive lymph nodes (`nodes`), tumor differentiation (`differ`), local extent of invasion (`extent`), and indicators for obstruction, perforation, and adherence to nearby organs.

Real data, rather than synthetic data alone, was used deliberately throughout: the literature anchors mostly demonstrate their methods on synthetic examples or a single toy application, and part of the point of this project is to see whether the qualitative conclusions replicate, and where the methods break, on a genuine dataset with realistic covariate structure, censoring, and sample sizes.

A key modeling choice is the cure/tail threshold τ (the point beyond which the distribution’s tail behavior is extrapolated). For recurrence, τ was set to $\tau_G + 2$ years (≈ 11.1 years), justified by external clinical evidence: a large pooled analysis (Pastorino et al., 2025, *JAMA Oncology*, $n = 35,213$ across 15 trials) shows colon-cancer recurrence risk falls below 0.5% by six years post-surgery. For death (all-cause mortality), no such disease-specific cure horizon exists, so a more generous $\tau_G + 10$ years was used. This tau-sensitivity turned out to matter substantively: an earlier, more naive placeholder tau ($1.5\times$ the generic horizon) produced only a borderline recurrence result ($p \approx 0.06\text{--}0.08$), while the clinically justified tau produced a clearly significant one ($p = 0.023$), illustrating that the choice of extrapolation horizon is not a nuisance parameter but a substantive, defensible modeling decision.

4 Phase 1: Baseline Replication of the Sufficient Follow-Up Tests

4.1 Method

The unconditional and categorical sufficient-follow-up tests of Yuen and Musta (2024) and Yuen et al. (2025) were applied to both endpoints of the `colon` data, using the authors’ own reference implementation, the `cureSFUTest` R package. Two categorical splits were tested: age (binned at the median) and sex.

4.2 Implementation issues found and fixed

Working with the real reference implementation surfaced concrete issues, all resolved and documented rather than worked around silently:

- `sfu.cov.test()` silently requires a numeric covariate matrix and raises a misleading error message otherwise.
- The package’s own bootstrap loop reads `.Random.seed` without checking it exists, which crashes on a fresh R session; fixed by explicitly calling `set.seed()` beforehand. This is a genuine bug in the published reference implementation, worth citing as such.
- The initial hand-written aggregation of *Method 1* (the intersection-union test) both (a) used the wrong, unconditional bandwidth instead of the paper’s undersmoothed conditional bandwidth, and (b) aggregated per-subgroup p -values with $\min(p_x)$ rather than the correct $\max(p_x)$ implied by the intersection-union logic (H_0 : insufficient for *some* x ; H_1 : sufficient for *all* x). Both were fixed – the corrected implementation now reads Method 1’s per-subgroup p -values directly off the same `sfu.cov.test()` call already used for Method 2, which internally always uses the correct undersmoothed bandwidth ($bw = \min(\tau n^{-7/30}, \tau/2)$). Getting the aggregation rule backwards would have silently flipped the recurrence-by-sex conclusion, so this correction mattered.

4.3 Results

Table 1: Unconditional sufficient-follow-up test results, by endpoint.

Endpoint	p -value	Conclusion
Recurrence	0.023	Sufficient follow-up
Death (all-cause)	0.66	Insufficient follow-up

Table 2: Categorical (Method 1, corrected) sufficient-follow-up test, per-subgroup p -values.

Split	Subgroup p -values	Method 1 ($\max p$) conclusion
Recurrence $\times$ sex	female 0.001, male 0.108	Not sufficient for <i>all</i> (male not significant)
Recurrence $\times$ age	0.015, 0.043	Sufficient for both age groups
Death $\times$ sex	0.328, 0.122	Insufficient
Death $\times$ age	0.400, 0.138	Insufficient

The headline pattern is stable across every subgroup and survives the bandwidth/aggregation bug fix unchanged: recurrence follow-up is adequate, while death (all-cause mortality) follow-up is not. This makes sense clinically: colon-cancer recurrence has a genuine, relatively short biological horizon, whereas all-cause death has no such horizon within a finite trial follow-up window.

The recurrence-by-sex split is the most interesting finding of this phase: female patients show clearly sufficient follow-up ($p = 0.001$), while male patients are borderline and not statistically distinguishable from insufficient ($p = 0.108$). Age, by contrast, does not meaningfully split the recurrence conclusion. This is exactly the kind of covariate-dependent effect that a purely unconditional test would miss.

A secondary finding, useful methodologically: [Yuen et al. \(2025\)](#)’s own paper flags that Method 2’s bootstrap-based worst-category selection can disagree with a naive largest- p -value ranking (“se-

lection instability”). This was independently reproduced on the death-by-age and death-by-sex splits, across two separate runs.

5 Phase 2: Extrapolation Estimator and the Curse of Dimensionality

5.1 Method

No existing R package implements the conditional, heavy-tailed correction estimator of [Escobar-Bach and Van Keilegom \(2019\)](#), so it was implemented from equations (2.6)–(2.8) of that paper. The base layer is a Beran (1981) kernel-weighted conditional Kaplan–Meier estimator (via `condSURV::Beran`, Epanechnikov kernel, bandwidth from `KernSmooth::dpik`). On top of this, a three-point ratio estimator gives the local tail index $\hat{\gamma}(x)$, and a two-point estimator gives the extrapolated cure probability $\hat{p}(x)$, using two geometrically-spaced tuning constants (y_1, y_2) selected over a grid via the paper’s “quadratic errors criterion.” This selection criterion was algebraically simplified to an equivalent, cheaper form: choosing the grid point whose $\hat{p}$ is closest to the mean of all grid $\hat{p}$ values, since $\arg \min_k \sum_j (P_k - P_j)^2 \Leftrightarrow \arg \min_k |P_k - \text{mean}(P)|$.

5.2 Validation and bugs found

The implementation was validated against synthetic data with known ground truth ($\gamma(x) = (x+1)/2$, a logistic cure-probability function $p(x)$, Pareto-tailed uncured survival times, and an artificially short follow-up window to force a genuine insufficient-follow-up bias). The first run produced impossible values ($\hat{p} > 1$), traced to a missing implementation detail: the source paper truncates $\hat{\gamma}(x)$ from below at 0.1 before it enters the $\hat{p}$ formula’s $y_1^{-1/\gamma}$ term. After adding this truncation, the estimator recovered sensible values close to ground truth, and moved the naive Beran estimate in the theoretically correct direction (upward, since naive Beran always underestimates the true non-cure probability under insufficient follow-up).

Applying the validated estimator to the real `colon` data surfaced a second, real-data-specific bug: on real data (a sparser effective local sample near the tail than the synthetic validation setup), $\hat{\gamma}(x)$ can swing very large (up to ≈ 40 at age 75), which sends $\hat{p}(x)$ outside $[0, 1]$ through the denominator $y_1^{-1/\gamma} - 1$ approaching zero. This was fixed by restricting the tuning grid to candidates whose implied $\hat{p}$ falls in the theoretically required range $[F_n(\tau_n), 1]$ before applying the closest-to-mean selection rule.

5.3 Result: recurrence vs. death, and triangulation with Phase 1

Applied to real `colon` data (age covariate, 9-point grid over ages 35–75), the corrected estimator satisfies $\hat{p}(x) \geq$ naive Beran everywhere, as theory requires, and the correction is markedly larger for death (mean gap 0.078) than for recurrence (mean gap 0.038). This is a genuine triangulation: two independent methods (the Phase 1 hypothesis test and the Phase 2 extrapolation estimator) *and* the source paper’s own colon application all agree that recurrence follow-up is close to adequate while death follow-up is not.

A sex-stratified check (splitting the same age-grid analysis by sex, to compare against the Phase 1 recurrence-by-sex finding) did *not* replicate cleanly: the mean correction gap was larger for females (0.029) than males (0.021), the opposite direction from the Phase 1 SFU test, and per-age gaps were noisy (e.g. one age point had no valid tuning pair at all). This is attributed to sample thinning: halving the sample per stratum shrinks the local effective sample size the Beran estimator sees

near the tail, making the tail-index estimation noticeably noisier. This non-replication is reported honestly rather than forced into agreement with Phase 1; the pooled recurrence-vs-death result remains the stronger, cleanly-triangulated finding.

5.4 Curse-of-dimensionality demonstration

The core methodological question the vacancy for this project is built around, statistical learning for high-dimensional covariates with theoretical guarantees, motivated a concrete, mechanism-level demonstration of *why* kernel-based conditional estimators like Beran’s break down as covariates accumulate, rather than a single abstract assertion that the curse of dimensionality applies. The Beran estimator was extended to a multivariate product-Epanechnikov kernel, and real `colon` recurrence covariates were added one at a time (age, then `+nodes`, `+differ`, `+extent`), evaluated at the componentwise-median “typical patient,” with a *fixed* per-dimension bandwidth (from `dpik`, or a Silverman-rule fallback for near-degenerate ordinal covariates) reused unchanged as dimensions were added. Holding bandwidths fixed as dimensions grow ensures that any degradation reflects dimensionality itself, not bandwidth retuning.

Table 3: Curse-of-dimensionality demonstration: effective sample size collapses as covariates accumulate.

Covariates	Kish ESS	naive p	corrected $\hat{p}$	$\hat{\gamma}$
age	134.8	0.508	0.546	0.936
+ nodes	36.6	0.359	0.395	1.214
+ differ	29.4	0.362	0.418	1.241
+ extent	18.4	0.495	0.496	0.100 (truncation floor)

The effective sample size (Kish’s ESS) collapses monotonically from 135 to 18 as four covariates are added with a fixed bandwidth. By four covariates, the tail-index estimator has visibly broken down: $\hat{\gamma}$ is pinned exactly at the 0.1 truncation floor introduced during validation, a clean numerical signature of outright estimation failure rather than a gradual, still-informative decline. This is concrete, auditable evidence of the curse-of-dimensionality problem, not an assertion.

A companion analysis multiplying dimensions differently (age, `+nodes`, `+sex`, `+differ`, evaluated on the death endpoint with $n = 888$) reinforced this at a different evaluation point and additionally showed that the failure mode is not uniform across covariate space. Repeating the correction-gap analysis across five evaluation points (young/old age $\times$ low/high `nodes` profiles, plus the median patient) revealed that the clean “gap shrinks monotonically with dimension” story from a single median-point run does *not* generalize: the mean correction gap trends down through three added covariates but ticks back up at the fourth, and $\hat{\gamma}$ floor-pinning is non-monotone across dimension.

The real driver turned out to be *local covariate-value rarity*, not raw dimension count: evaluation points with unusual `nodes` values (e.g. `nodes`= 8, versus a typical range of 1–5) collapse to an effective sample size of only 7–11 already at two covariates, and show $\hat{\gamma}$ bouncing erratically as further dimensions are added. Central, typical points (the median patient, low-nodes patients) stay comparatively stable (ESS 60–160) even at four covariates.

A subsequent $B = 150$ bootstrap check at three representative points (the median patient, a young/high-nodes patient, and an old/high-nodes patient), repeated across all four dimensions, confirmed this is a resampling-stable pattern rather than a single-sample artifact: for the typical/median patient, $\hat{\gamma}$ floor-pinning rises from 5% of bootstrap resamples at one covariate to 63–69% at three to four covariates, a systematic, dimensionality-driven collapse, while atypical

high-nodes points are already unstable at a single covariate and simply stay noisy rather than worsening further, a small-sample fragility that dimensionality compounds rather than causes.

This yields a sharper conclusion than a single blanket curse-of-dimensionality statement: there are *two separable failure modes*: systematic, genuine dimensionality collapse for typical patients, and pre-existing small-sample fragility for atypical, rare covariate profiles that dimensionality merely compounds.

5.5 Comparison against a gradient-boosted alternative (gbex)

To test whether a modern high-dimensional tail-regression tool resolves either failure mode, **gbex** (Velthoen et al., 2021) (a gradient-boosted generalized Pareto regression method, co-authored by one of this project’s target supervisors) was installed from source (worked around a GitHub API rate limit by downloading the tarball directly via codeload.github.com rather than `remotes::install_github`) along with its undeclared dependencies (POT, treeClust).

Getting **gbex** to run correctly on this data required fixing several real usage bugs: the function signature does not accept a `status/threshold/maxit/stopcrit` interface (an initial implementation attempt assumed it did); `predict()` does not support `type="response"` and instead returns a data frame with columns `s` (scale) and `g` (shape); and covariates must be passed as a proper data frame with real column names, not a bare matrix, or `predict.gbex` silently mismatches columns internally.

A more fundamental limitation was then found: **gbex** has no built-in notion of censoring, so it can only be fit on *uncensored* exceedances above a threshold. Counting usable points across candidate thresholds showed this is severely limiting on this dataset: for recurrence (whose follow-up is already adequate per Phase 1, so there is little tail-exceedance signal left to model), the usable count drops from 32 points at the median threshold to just 1 at the 75th/90th percentile; for death, from 33 at the median threshold to 4 at the 90th percentile. Given this, the comparison was deliberately scoped down to the death endpoint at a low (50th-percentile) threshold, where 33 uncensored exceedances are available.

On this constrained but usable setup, **gbex** produced a tail-index estimate $\hat{\gamma} \approx 0.005$ (near-exponential) that was essentially *constant* across every evaluation point and every covariate dimension tested. Rather than accepting this as genuine stability, the flat result was investigated further. The root cause: with only 33 observations, **gbex**’s internal unconditional maximum-likelihood initial guess for γ came back negative (nonsensical for a generalized Pareto tail), silently triggering a fixed default initialization ($\gamma = 0.01$); from that fixed starting point, the boosting shrinkage is tiny ($\lambda \approx 0.007$), so the reported estimate barely moves regardless of the covariates supplied. A 30-repetition bootstrap confirmed the estimate is frozen under resampling too (SD = 0.00012), i.e. the estimator is not responding to the data at all, not merely stable.

This is an important negative finding, arguably more important than a clean positive result would have been: at this sample size, neither method reliably recovers patient-level tail heterogeneity, but the two methods fail in qualitatively different ways. The Beran-based estimator fails *visibly* – explicit truncation-floor pinning is an auditable, detectable failure signature. **gbex** fails *silently*: it produces what looks like a converged, stable estimate, but the stability is an artifact of a fallback initialization rather than genuine convergence. The silent failure mode is arguably the more practically dangerous one, since it could mislead a practitioner who does not inspect the initialization behavior.

Separately, **gbex** was also applied to the recurrence endpoint using all ten available real covariates simultaneously (age, sex, obstruction, perforation, adherence, differentiation, extent, surgery type, nodes, and a `nodes $\geq$ 4` indicator) on 223 uncensored exceedances, explicitly as a demonstration

of covariate-scaling for the *scale* parameter alone, not a claim of solving the censoring/cure problem directly (this remains the open methodological gap the target PhD project is meant to address). Here **gbex** performed well: the estimated scale parameter varied substantially and meaningfully across patients (0.31–3.13), driven mainly by age, number of positive nodes, differentiation, and obstruction, and the full ten-covariate model outperformed an age-only model on deviance (1.221 vs. 1.280). The estimated shape (tail index), however, stayed pinned near zero for every patient regardless of covariates – consistent with the unconditional estimate, which was actually negative before **gbex**’s positivity constraint clamped it. This suggests that, among patients who do recur, the raw recurrence-time distribution shows little evidence of genuine heavy-tailedness; the insufficient-follow-up problem in this data more plausibly comes from the cure/mixture structure itself (a real never-recur subpopulation) than from a heavy-tailed susceptible population. This is a useful methodological point in its own right: it argues for the mixture-cure framing used throughout this project over a naive heavy-tail-only model, while also demonstrating **gbex**’s genuine value on the scale parameter specifically.

6 Phase 5: Simulation Study

Phase 5 was prioritized over the originally planned Phase 3 (a continuous-covariate sufficient-follow-up test) and the full Phase 4 (a comprehensive per-estimator uncertainty-quantification framework) because it reused already-validated code at low marginal cost and produced results with direct interpretive value. Phase 3 was explicitly scoped as a stretch or sketch item from the outset, and the full Phase 4 was judged to be substantial, open-ended work relative to the remaining time before the application deadline.

6.1 Part A: Bias–variance tradeoff as follow-up length varies

Using the validated Phase 2 estimator on $N = 2000$ synthetic patients (true $p(x_0 = 0.5) = 0.5987$), the censoring horizon τ_c was swept over $\{4, 6, 8, 12, 20, 40\}$, 30 replications per value.

Table 4: Bias and MSE of naive Beran vs. EBVK-corrected estimator as follow-up lengthens (τ_c).

τ_c	Bias (naive)	Bias (corrected)	MSE (naive)	MSE (corrected)
4 (severe)	−0.095	−0.038	0.0119	0.0038
6	−0.038	+0.018	0.0044	0.0039
8	−0.031	+0.012	0.0042	0.0053
12	−0.024	−0.003	0.0027	0.0027
20	−0.002	+0.009	0.0015	0.0018
40 (near-sufficient)	+0.003	+0.011	0.0022	0.0026

This demonstrates the bias-variance tradeoff the correction is theoretically expected to exhibit, though the pattern is not perfectly monotonic. At severe insufficiency ($\tau_c = 4$), the corrected estimator more than halves the bias and wins decisively on MSE (0.0038 vs. 0.0119), exactly where correcting matters most, and it still wins narrowly at $\tau_c = 6$ (0.0039 vs. 0.0044). By $\tau_c = 8$ the correction’s MSE advantage has already flipped (0.0053 vs. 0.0042), before the two estimators tie again at $\tau_c = 12$ and naive Beran pulls ahead from $\tau_c = 20$ onward. With only 30 replications per τ_c , the dip at $\tau_c = 8$ is plausibly Monte Carlo noise rather than a genuine non-monotonicity, and it does not change the overall picture: as follow-up approaches sufficiency, naive Beran’s own

bias shrinks toward zero on its own, so the correction’s advantage fades and eventually reverses, since the correction adds variance without a meaningful bias left to remove. This confirms, on a controlled synthetic sweep rather than a single point estimate, that the correction should be applied when follow-up is insufficient and can be safely skipped once it is close to adequate.

6.2 Part B: Method 1 vs. Method 2 as the number of covariate categories grows

Using the real `cureSFUTest` package, the number of covariate categories K was swept over 2–5, with a fixed total sample size $n = 400$ split evenly across categories (so per-category n shrinks with K : 200, 133, 100, 80).

An initial pass mislabeled which simulated scenario corresponds to which error type; this was caught and corrected before reporting results below. In the “all categories sufficient” scenario the ground truth is H_1 , so *failing* to detect it is a Type II error, and the complement is the test’s power to confirm true full sufficiency. In the “one category insufficient” scenario the ground truth is H_0 , so wrongly declaring “all sufficient” is the actual Type I error.

Table 5: Method 1 (intersection-union test) power and Type I error, and Method 2 selection accuracy, as K grows.

K	$n/\text{category}$	Power (confirm true sufficiency)	Type I error (false “all clear”)	Method 2 selection accuracy
2	200	0.60	0.067	1.00
3	133	0.43	0.00	0.967
4	100	0.27	0.00	0.967
5	80	0.30	0.00	1.00

This produces a complementary-strengths finding. Method 1’s Type I error (falsely declaring “sufficient follow-up for all subgroups” when at least one subgroup truly is not) stays at or near zero across every K tested, and if anything the test becomes *more* conservative as K grows: the intersection-union test essentially never gives false reassurance. This safety has a real cost, however: power to positively *confirm* genuine full sufficiency collapses from 60% at $K = 2$ to roughly 27–30% at $K \geq 4$, because every one of K increasingly noisy, shrinking-sample-size, per-category tests must simultaneously clear the 5% significance bar. Method 2’s bootstrap-based diagnostic selection of *which* category is problematic, by contrast, stays reliable (97–100% accurate) throughout, regardless of K . The practical implication: Method 1 should be trusted for its conservatism (it will not falsely clear a real problem) but not relied upon to certify “everything is fine” once several covariate categories are involved; Method 2 remains the right tool for localizing where a problem is once one is suspected.

7 Discussion: What This Project Demonstrates

Across all five completed phases, three findings stand out as the strongest material for further development:

1. **Triangulated main finding (Phases 1 and 2).** Two independent methods, a hypothesis test and an extrapolation estimator, and the original source paper’s own real-data application all agree: colon-cancer recurrence follow-up in this dataset is close to sufficient, while all-cause death follow-up is not. This agreement across independently implemented methods is strong evidence that the qualitative conclusion is real, not an artifact of one estimator.

2. **Mechanism-level curse-of-dimensionality evidence (Phase 2).** Rather than asserting that kernel-based conditional cure estimators degrade with more covariates, this project traces the exact numerical mechanism: effective sample size collapse, and a clean truncation-floor signature marking the point of estimator breakdown. It further separates this into two distinct failure modes, systematic dimensionality collapse for typical patients, and pre-existing small-sample fragility for atypical ones, and shows that a modern gradient-boosted alternative (`gbex`) does not straightforwardly solve the problem: at low sample sizes it fails *silently*, via a fallback initialization, which is arguably more dangerous in practice than the Beran estimator’s visible truncation-floor failure.
3. **Bias–variance tradeoff (Phase 5, Part A).** A controlled synthetic sweep over follow-up length reproduces, with a full multi-point curve rather than a single spot check, the theoretical bias–variance tradeoff the correction method is designed to exhibit: large, decisive gains when follow-up is severely insufficient, and a cost in variance once follow-up is already close to adequate, with one likely-noise dip at $\tau_c = 8$ noted rather than smoothed over.

An honest secondary finding worth naming explicitly is the sex-stratified non-replication in Phase 2 (Section 4.3): the Phase 1 hypothesis test and the Phase 2 extrapolation estimator disagree on the direction of the recurrence-by-sex effect once the sample is split by sex, most plausibly because of sample-size thinning rather than a genuine methodological contradiction. This is reported as an honest limitation rather than adjusted to agree with the headline result.

Two concrete implementation bugs in the public `cureSFUTest` reference package, and several more in this project’s own from-scratch code (a missing tail-index truncation, an out-of-range tuning-grid selection, an intersection-union aggregation-rule error, an incorrect Type I/II error labeling in simulation code), were found and corrected during the course of this work. All are documented above rather than silently fixed, since correctly diagnosing and fixing implementation bugs in both a published reference package and one’s own code is itself part of the applied statistical skill this project is meant to demonstrate.

8 Future Work

Two components of the original six-phase plan were deliberately deferred, and are named here as concrete open problems rather than treated as gaps in the current work.

A continuous-covariate sufficient-follow-up test. The categorical test of [Yuen et al. \(2025\)](#) requires discretizing continuous covariates (as was done here for age) into bins, which discards information and makes the Phase 1 power problem (Section 6, Part B) worse as more categories are used. A natural extension combines Yuen & Musta’s smoothed-Grenander boundary estimator with the Beran-type conditional weighting used in Phase 2, replacing discrete subsampling by continuous covariate weighting. This does not require a full asymptotic theory to be useful as a first step: a working numerical version, paired with an honest statement of exactly which asymptotic argument (e.g. the limiting distribution of the smoothed-Grenander statistic under continuous local weighting) remains open, is itself a meaningful contribution and a natural bridge into doctoral-level theoretical work on this problem.

Per-estimator uncertainty quantification, done correctly. The tests used in Phases 1 and 5(B) come with their own bootstrap-based p -values, but the Phase 2 extrapolation estimator ($\hat{p}(x)$, $\hat{\gamma}(x)$) currently has no confidence intervals at all. A methodologically correct treatment requires different resampling schemes for different estimator types: a naive bootstrap is appropriate for

smooth functionals (such as the Xie et al. and Escobar-Bach & Van Keilegom estimators), while a smoothed bootstrap is required for the shape-constrained, Grenander-based tests, whose limiting distributions are non-standard. Building this properly, per estimator type, is substantial methodological work in its own right and is left as future work rather than attempted with a one-size-fits-all interval that would likely be invalid for at least one of the estimator types involved.

Resolving the silent gbex failure mode. The Phase 2 comparison surfaced a specific, fixable issue: **gbex**’s reliance on an unconditional MLE initial guess for the tail index becomes unreliable at small uncensored sample sizes, silently falling back to a fixed default. A natural methodological extension, and a direct answer to the vacancy’s stated interest in high-dimensional covariates with theoretical guarantees, is to extend **gbex** or a similar gradient-boosted tail-regression framework to handle censoring natively (e.g. via a censored generalized Pareto likelihood), rather than restricting it to uncensored exceedances as done here. This would substantially increase the usable sample size for the recurrence endpoint in particular, where follow-up is already adequate and censored near-threshold observations currently carry no weight in the tail-regression step.

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